DIFFERENTIAL COHOMOLOGY SEMINAR 2

TALK BY MATTHIAS FRERICHS

In this lecture we want to learn the basics of ∞-category theory. For the ∞-categorical background, we broadly follow [Gro10] and a little [Lur09]. This talk assumes familiarity with the language of model categories, see [Hir03].

1. Basics on $(\infty, 1)$-categories

There are a pletora of models for ∞-categories, here we focus on two, namely simplicial sets satisfying a filling condition, and categories enriched over simplicial sets.

1.1. Simplicial sets.

We begin by recalling some definitions about simplicial sets. Denote by $\mathcal{POS}\mathrm{et}$ the category of finite, linearly ordered sets, and $\mathbf{\Delta} \subseteq \mathcal{POS}\mathrm{et}$ the full sub-category of sets $[n] = \{0, \cdots, n\}$ (with natural ordering), for every natural number n. Given a category $\mathcal{C}$, a *simplicial object* is a functor $S : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$, the value at $[n]$ is denoted by S_n. Simplicial objects in $\mathcal{C} = \mathcal{S}\mathrm{et}$ are called *simplicial sets*, the category of simplicial sets is depicted as $\mathrm{s}\mathcal{S}\mathrm{et}$.

Definition 1.1. Let $0 \leq i \leq n$. Given a simplicial object S, Then

(1) The *i-face map* $d_i : S_n \to S_{n-1}$ is the one induced by the unique injective, monotone function $f : [n-1] \to [n]$ such that i doesn't belong in the image of f.
(2) The *i-degeneracy map* $s_j : S_n \to S_{n+1}$ is the one induced by the unique surjective, monotone function $f : [n+1] \to [n]$ such that $f(i) = f(i+1) = i$.

We also use the following special notation:

Notation 1.2. 0-simplices are called *vertices*, while 1-simplices are called *edges*. The 1-face, resp. 0-face, map $S_1 \to S_0$ is called *source*, resp. *target*, and denoted s, resp. t. Given an edge f, we write $f : X \to Y$ to mean X is the source and Y the target.

Face and generacy maps (and objects S_n, for every n) fully determine the simplicial object S, see [GJ99, I.4]. By only depicting face and degeneracy maps, we can represent a simplicial object as follows:

$$S_0 \;\substack{\longleftarrow\\ \longleftarrow\\ \longrightarrow}\; S_1 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longrightarrow\\ \longrightarrow}\; S_2 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots \tag{1.3}$$

Example 1.4. Given a natural number n, denote by Δ^n the simplicial set represented by $[n]$. In particular, $(\Delta^n)_k$ is the set of monotone functions $[k] \to [n]$. By Yoneda, given a simplicial set S, we can (and will) identify the set S_n with $\mathrm{Hom}_{\mathrm{s}\mathcal{S}\mathrm{et}}(\Delta^n, S)$. An element $\in S_n$ is called a *n-simplex*.

Example 1.5. Given a natural number n, denote by $\partial\Delta^n \subseteq \Delta^n$ the *boundary* of Δ^n, defined as follows: $f : [k] \to [n]$ in $(\Delta^n)_k$ belongs to $(\partial\Delta^n)_k$ if f is non-surjective.

Example 1.6. Given $0 \leq i \leq n$, denote by $\Lambda^n_i \subseteq \Delta^n$ the *i-horn*, defined as follows: $f : [k] \to [n]$ in $(\Delta^n)_k$ belongs to $(\Lambda^n_i)_k$ if $\{i\} \cup f([k]) \neq [n]$. *Right*, resp. *left*, horns are all horns Λ^n_i with $0 < i \leq n$, resp. $0 \leq i < n$. *Inner* horns are all horns Λ^n_i with $0 < i < n$.

1.2. ∞-Categories.

Definition 1.7. Let $\mathcal{C}$ be a category, $f : X \to Y$ has the *right lifting property* against $i : A \to B$ if for every solid diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "f"] \\ B \arrow[r] \arrow[ur, dashed, "\exists"] & Y \end{tikzcd} \tag{1.8}$$

Date: 06.05.2025 & 13.05.2025.

there is a dashed arrow making the diagram commute.

Definition 1.9. A (*inner/right/left*) *fibration* is a morphism of simplicial sets $f : X \to Y$ that has the right lifting property against all (inner/right/left) horn inclusions. A simplicial set X is called an ∞-*category*, resp. ∞-*groupoid*, if the terminal map $X \to \Delta^0$ is an inner fibration, resp. a fibration.

Example 1.10. The nerve of a category, resp. groupoid, is an ∞-category, resp. ∞-groupoid.

We recall briefly the construction of the *homotopy category* $\mathrm{h}S$ of a ∞-category S, details can be found in [Lur09, Lur26e]. The objects of $\mathrm{h}S$ are the vertices of S. Given a vertex X, denote by id_X the image of X under $s_0 : S_0 \to S_1$. Given two vertices X, Y and edges $f, g : X \to Y$, we say f is *left homotopic* to g if there is a 2-simplex with the following boundary:

$$\begin{tikzcd} & Y \arrow[dr, "\mathrm{id}_Y"] & \\ X \arrow[ur, "f"] \arrow[rr, "g"'] & & Y \end{tikzcd} \tag{1.11}$$

For S an ∞-category, being left homotopic is an equivalence relation on the set of edges $X \to Y$. Denote by $\mathrm{Hom}_{\mathrm{h}S}(X, Y)$ the set of edges $X \to Y$ modulo being left homotopic. Given an edge f, we write $[f]$ for its left homotopy class. Two edges $f : X \to Y$ and $g : Y \to Z$ determine a map $\Phi : \Lambda^2_1 \to S$. By definition, there is a 2-simplex σ such that $\sigma|_{\Lambda^2_1} = \Phi$, then

$$[g] \circ [f] := [d_1(\sigma)] \tag{1.12}$$

This composition is well-defined, associative, and unital (with $[\mathrm{id}_X]$ acting as the identity of X).

Definition 1.13. Let S be a ∞-category, an edge f is an *equivalence* if $[f]$ is an isomorphism in $\mathrm{h}S$.

Theorem 1.14 ([Lur26b])**.** *A ∞-category S is a ∞-groupoid if and only if $\mathrm{h}S$ is a groupoid.*

1.3. Simplicial categories.

Definition 1.15. A *simplicial category* is a category enriched over simplicial sets. Denote by $\mathrm{s}\mathcal{C}\mathrm{at}$ the category of simplicial categories.

Recall that $\mathrm{s}\mathcal{S}\mathrm{et}$ is self-enriched as follows:

Definition 1.16. Given two simplicial sets X, Y, the *hom-simplicial set* $\mathrm{Hom}^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(X, Y)$ is defined as follows: A n-simplex consists of a map of simplicial sets $\Delta^n \times X \to Y$.

Lemma 1.17. *If X is a ∞-groupoid, then $\mathrm{Hom}^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B, X)$ is a ∞-groupoid, for every simplicial set B.*

Proof. Corollary of [Lur26a] by taking f the terminal map $X \to \Delta^0$ and i the initial map $\emptyset \to B$. □

Definition 1.18. Let $(P, \leq)$ be a partially ordered set, denote by $\mathrm{Path}(P)$ the following simplicial category:

1. Objects are the elements of P.
2. Given $x, y \in P$, let $[x, y] := \{z \in P | x \leq z \leq y\}$ and $P_{x,y}$ the set of sub-sets $S \subseteq [x, y]$ that are finite, linearly ordered, and contain $\{x, y\}$. We order $P_{x,y}$ by reverse inclusion ($S \leq T$ if $S \supseteq T$). Finally, let $\mathrm{Hom}_{\mathrm{Path}(P)}(x, y)$ be the nerve of $(P_{x,y}, \supseteq)$.
3. Given $x, y, z \in P$, composition

$$\mathrm{Hom}_{\mathrm{Path}(P)}(x, y) \times \mathrm{Hom}_{\mathrm{Path}(P)}(y, z) \longrightarrow \mathrm{Hom}_{\mathrm{Path}(P)}(x, z) \tag{1.19}$$

is the nerve of the union map $\cup : P_{x,y} \times P_{y,z} \to P_{x,z}$.

Given a monotone map $f : P \to Q$, let $\mathrm{Path}(f) : \mathrm{Path}(P) \to \mathrm{Path}(Q)$ denote the following functor:

1. On objects, it is simply f.
2. Given $x, y \in P$, define $\mathrm{Path}(f)$ as the nerve of the functor $f_* : P_{x,y} \to Q_{f(x),f(y)}$ given by taking direct images (i. e. $f_*(S) = f(S)$).

Restricting to $\mathbf{\Delta}$, we obtain a cosimplicial simplicial category $\mathrm{Path} : \mathbf{\Delta} \to \mathrm{s}\mathcal{C}\mathrm{at}$.

Definition 1.20. Given a simplicial category $\mathcal{C}$, its *simplicial nerve* is the simplicial set $\mathrm{N}_{\Delta}\mathcal{C}$ defined as follows: A n-simplex of $\mathrm{N}_{\Delta}\mathcal{C}$ consists of a functor $\mathrm{Path}_n := \mathrm{Path}(\langle n \rangle) \to \mathcal{C}$.

Remark 1.21. Path_0 and Path_1 are the categories $\langle 0\rangle$ and $\langle 1\rangle$ viewed as simplicially enriched categories with discrete hom-simplicial sets, therefore a functor $\mathrm{Path}_0 \to \mathcal{C}$, resp. $\mathrm{Path}_1 \to \mathcal{C}$, is determined by a single object, resp. 1-morphism. Path_2 admits the following description:

(1) Its set of objects is $\langle 2\rangle$.
(2) $P_{0,1}$ and $P_{1,2}$ are both singletons, hence $\mathrm{Hom}_{\mathrm{Path}_2}(0,1) \cong \mathrm{Hom}_{\mathrm{Path}_2}(1,2) \cong \Delta^0$.
(3) $P_{0,2} = \{[0,2] \leq \{0,2\}\}$ (recall that $P_{i,j}$ is ordered by reversed inclusion), so $\mathrm{Hom}_{\mathrm{Path}_2}(0,2) \cong \Delta^1$. Moreover, $[0,2]$ is the composition $\{1,2\} \circ \{0,1\}$.

We can conclude that a functor $\mathrm{Path}_2 \to \mathcal{C}$ is determined by the following data: A triple of 1-morphisms $f : x \to y, g : y \to z$, and $h : x \to y$, together with a 1-simplex $g \circ f \Rightarrow h$ in $\mathrm{Hom}_{\mathcal{C}}(x,z)$.

Definition 1.22. Denote also by Path the functor $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{C}\mathrm{at}$ obtained by left Kan extension of Path : $\mathbf{\Delta} \to \mathrm{s}\mathcal{C}\mathrm{at}$.

Theorem 1.23. *The functor* Path *is left adjoint to* N_Δ *and form a Quillen equivalence between the* Dwyer-Kan-Bergner *model structure on* $\mathrm{s}\mathcal{C}\mathrm{at}$ *and the* Joyal *model structure on* $\mathrm{s}\mathcal{S}\mathrm{et}$. *The fibrant objects in the* $\mathrm{s}\mathcal{S}\mathrm{et}_{\mathrm{Joyal}}$*, resp.* $\mathrm{s}\mathcal{C}\mathrm{at}_{\mathrm{DKB}}$*, are* ∞*-categories, resp. locally fibrant categories. In particular,* $\mathrm{N}_\Delta\mathcal{C}$ *is a* ∞*-category if* $\mathcal{C}$ *is locally fibrant.*

By Equation (1.17), we know that the full sub-category $\infty\mathcal{G}\mathrm{rpd} \subseteq \mathrm{s}\mathcal{S}\mathrm{et}$ of ∞-groupoids is self-enriched, therefore we give the following definition.

Definition 1.24. Denote by $\mathcal{S} := \mathrm{N}_\Delta(\infty\mathcal{G}\mathrm{rpd})$ the ∞*-category of* ∞*-groupoids.*

Remark 1.25. In view of [Lur26c], we will also use the term *space* to mean ∞-groupoid.

2. Basic constructions

Recall that $\mathrm{s}\mathcal{S}\mathrm{et}$ and $\infty\mathcal{G}\mathrm{rpd}$ are self-enriched. Similar to Equation (1.17), we have the following:

Lemma 2.1 ([Lur26d])**.** *If* X *is a* ∞*-category, then* $\mathrm{Hom}^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B,X)$ *is a* ∞*-category, for every simplicial set* B*.*

Definition 2.2. Given a ∞-category X and simplicial set B, denote by $\mathcal{F}\mathrm{un}(B,X)$ the ∞*-category of functors* $\mathrm{Hom}^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B,X)$.

Definition 2.3. Denote by op the following endo-functor of $\mathbf{\Delta}$:

(1) On objects, $\mathrm{op}(\langle n\rangle) = \langle n\rangle$.
(2) Given $f : \langle k\rangle \to \langle n\rangle$, let $\mathrm{op}(f) : \langle k\rangle \to \langle n\rangle$ be the function $\mathrm{op}(f)(i) = n - f(k-i)$.

Given a simplicial set X, denote by X^{op} the pre-composition of X with op.

Lemma 2.4. *If* X *is a* ∞*-category,* X^{op} *is also an* ∞*-category.*

Definition 2.5. Given a ∞-category X, we call X^{op} the *opposite* ∞*-category.*

Slice ∞-categories?

3. Accessible and Presentable Categories

In general, a limit, resp. colimit, preserving functor need not have a left, resp. right, adjoint. Here we wish to introduce a rather large class of ∞-categories for which the previous statement holds. We begin by recalling the definition of locally presentable 1-categories. Let κ denote a regular cardinal.

Definition 3.1. A category $\mathcal{I}$ is κ*-filtered* if, for every $\mathcal{J}$ with $< \kappa$ many morphisms, every diagram $\mathcal{J} \to \mathcal{I}$ has a cocone. A functor $F : \mathcal{C} \to \mathcal{D}$ is κ*-accessible* if it preserves κ-filtered colimits. Given a category $\mathcal{C}$, an object X is κ*-compact* if $\mathrm{Hom}_{\mathcal{C}}(X,-) : \mathcal{C} \to \mathcal{S}\mathrm{et}$ is κ-accessible.

Definition 3.2. A category $\mathcal{C}$ is κ*-accessible* if there exists a set $S \subseteq \mathcal{C}_0$ of κ-compact objects that generate $\mathcal{C}$ under κ-filtered colimits. A category is *accessible* if it is κ-accessible, for some regular cardinal κ.

Definition 3.3. A category $\mathcal{C}$ that is accessible and cocomplete is called *locally presentable.*

Theorem 3.4. *Let* $\mathcal{C}$ *be a category, then* $\mathcal{C}$ *is locally presentable if and only if there exists a small category* S *such that the induced functor* $\mathcal{C} \to \mathcal{P}(S)$ *is a fully faithful, accessible right adjoint.*

Theorem 3.5. *Let $\mathcal{C}, \mathcal{D}$ be locally presentable categories. A functor $F\colon \mathcal{C} \to \mathcal{D}$ is a left, resp. right, adjoint if and only if it preserves colimits, resp. it preserves limits and is accessible.*

Definition 3.6. Let X be a ∞-category. The *∞-category of pre-sheaves of spaces* is defined as $\mathcal{P}(X) := \mathrm{Hom}_{\mathrm{sSet}}(X^{\mathrm{op}}, \mathcal{S})$.

Theorem 3.7 (Yoneda)**.** *Given a ∞-category X, there is a fully faithful functor $y : X \to \mathcal{P}(X)$, called the* Yoneda embedding*, such that: Given a cocomplete ∞-category Y, pre-composition by y induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{P}(X), Y) \longrightarrow \mathcal{F}\mathrm{un}(X, Y) \tag{3.8}$$

where $\mathcal{F}\mathrm{un}^{\mathrm{L}}$ denotes the category of colimit preserving functors.

The definition of accessible category transfers directly to the ∞-categorical setting.

Theorem 3.9. *A ∞-category X is locally presentable (cocomplete and accessible) if and only if there is a small sub-∞-category S such that $X \to \mathcal{F}\mathrm{un}(S^{\mathrm{op}}, \mathcal{S})$ is a fully faithful, accessible right adjoint.*

Remark 3.10. In view of Equation (3.9), one can define a category to be *locally presentable* if it is the accessible right localization of a pre-sheaf category for some small ∞-category S. In particular, every pre-sheaf category is locally presentable.

Theorem 3.11. *Let X, Y be presentable ∞-categories, then a functor $f : X \to Y$ is a left, resp. right, adjoint if and only if it preserves colimits, resp. it preserves limits and is accessible.*

4. Stable ∞-Categories and Spectra

We now use the ∞-categorical framework to study spectra. Here are some result that justify the study of spectra.

Examples of stable phenomena? Freudenthal's suspension theorem, Alexander duality of manifolds, etc.

Definition 4.1. Let $\mathcal{C}$ be an ∞-category, an object 0 that is both initial and terminal is called *zero object*. A category $\mathcal{C}$ with a zero object is called a *pointed category*.

Example 4.2. Let $1 \in \mathcal{C}$ be a terminal object, then the identity of 1 is the zero object in the slice category $\mathcal{C}_{1/}$ of objects under 1. In particular, the category $\mathcal{S}_* = \mathcal{S}_{*/}$ of pointed spaces is pointed.

Proposition 4.3. *Let $\mathcal{D}$ be a pointed ∞-category. Evaluation at the 0-sphere S^0 induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}_*, \mathcal{C}) \longrightarrow \mathcal{C} \tag{4.4}$$

We now introduce the notion of a triangle.

Definition 4.5. Let $\mathcal{C}$ be a pointed ∞-category. A *triangle* in $\mathcal{C}$ is a commutative diagram of the form

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Y \arrow[d] \\ 0 \arrow[r] & Z \end{tikzcd} \tag{4.6}$$

A triangle is *exact*, resp. *coexact*, if it is a pullback, resp. pushout, square.

Definition 4.7. Let $\mathcal{C}$ be a pointed ∞-category. Denote by $\mathcal{C}^{\Sigma}$, resp. $\mathcal{C}^{\Omega}$, the full sub-category of $\mathcal{F}\mathrm{un}(\Delta^1 \times \Delta^1, \mathcal{C})$ of coexact, resp. exact, triangles of the form

$$\begin{tikzcd} X \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & Y \end{tikzcd}, \tag{4.8}$$

If $\mathcal{C}$ is finitely cocomplete, resp. complete, for every object X, resp. Y, there is a contractible space of coexact, resp. exact, triangles as Equation (4.8). In particular, $\mathcal{C} \simeq \mathcal{C}^{\Sigma}$ and $\mathcal{C} \simeq \mathcal{C}^{\Omega}$.

Proposition 4.9. *If $\mathcal{C}$ is a finitely complete and cocomplete pointed ∞-category, define the following functors Then the functors*

$$\Sigma : \mathcal{C} \longrightarrow \mathcal{C}^{\Sigma} \xrightarrow{\mathrm{ev}_{(1,1)}} \mathcal{C} \qquad\qquad \Omega : \mathcal{C} \longrightarrow \mathcal{C}^{\Omega} \xrightarrow{\mathrm{ev}_{(0,0)}} \mathcal{C} \tag{4.10}$$

are adjoint (Σ is left adjoint to Ω).

Theorem 4.11. *Let $\mathcal{C}$ be a finitely bicomplete pointed ∞-category. The following are equivalent:*

(1) A triangle is exact if and only if it is coexact.
(2) (Σ, Ω) is an adjoint equivalence.
(3) A commutative square is a pullback if and only if it is a pushout.

Definition 4.12. A finite bicomplete pointed ∞-category $\mathcal{C}$ satisfying any of the equivalent conditions in Equation (4.11) is called *stable*.

If $\mathbb{A}$ denotes a nice abelian category, there is a *derived ∞-category* of $\mathbb{A}$, denoted $\mathcal{D}(\mathbb{A})$ such that $\mathrm{h}\mathcal{D}(\mathbb{A}) = D(\mathbb{A})$ is the ordinary derived category of $\mathbb{A}$, i.e. the localization of chain complex at quasi-isomorphisms. From homological algebra, it is known that $D(\mathbb{A})$ underlies the structure of a triangulated category, which turns out to be the 1-categorical reflection of $\mathcal{D}(\mathbb{A})$ being stable.

Proposition 4.13 ([Lur17, 3.11])**.** *If $\mathcal{C}$ is a stable ∞-category, then $\mathrm{h}\mathcal{C}$ has a canonical structure of a triangulated category.*

To construct the stabilization of a pointed ∞-category, there are several approaches, such as reduced excisive functors on $\mathcal{S}_*^{\mathrm{fin}}$, the category of pointed, finite spaces, see [Lur17, 1.4.2.8]. Here we consider the more explicit approach using spectrum objects.

Definition 4.14. Let $\mathcal{C}$ be a pointed ∞-category. A *pre-spectrum object in C* consists of a functor $E : \mathbb{Z} \times \mathbb{Z} \to \mathcal{C}$ such that $E(n, m) \simeq 0$, for all $n \neq m$. Denote by $\mathcal{PS}\mathrm{p}(\mathcal{C})$ the category of pre-spectrum objects. The functor $\Omega^{\infty - n} : \mathcal{PS}\mathrm{p}(\mathcal{C}) \to \mathcal{C}$ is defined as evaluation at (n, n).

For every n, the diagram

$$\begin{tikzcd} E(n,n) \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & E(n+1, n+1) \end{tikzcd} \tag{4.15}$$

determines a pair of adjoint morphisms

$$\alpha_n : \Sigma E(n, n) \to E(n + 1, n + 1), \qquad \beta_n : E(n, n) \to \Omega E(n + 1, n + 1)$$

Definition 4.16. Let $\mathcal{C}$ be a pointed ∞-category. A *spectrum object in* $\mathcal{C}$ consists of a pre-spectrum object E such that β_n is an equivalence, for all n. Denote by $\mathcal{S}\mathrm{p}(\mathcal{C}) \subseteq \mathcal{PS}\mathrm{p}(\mathcal{C})$ the full sub-category of spectrum objects.

Theorem 4.17. *Let $\mathcal{C}$ be a presentable pointed ∞-category, then $\Omega^{\infty - n} : \mathcal{S}\mathrm{p}(\mathcal{C}) \to \mathcal{C}$ admits a left adjoint $\Sigma^{\infty - n} : \mathcal{C} \to \mathcal{S}\mathrm{p}(\mathcal{C})$, for every n.*

In particular, $\Sigma^\infty : \mathcal{C} \to \mathcal{S}\mathrm{p}(\mathcal{C})$ has the following universal property.

Theorem 4.18. *Let $\mathcal{C}$ be a presentable pointed ∞-category. Given a stable ∞-category $\mathcal{D}$, pre-composition by Σ^∞ induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}\mathrm{p}(\mathcal{C}), \mathcal{D}) \longrightarrow \mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{C}, \mathcal{D}) \tag{4.19}$$

In particular, for $\mathcal{C} = \mathcal{S}_*$, evaluation at the sphere spectrum $\mathbb{S} = \Sigma^\infty S^0$ *induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}\mathrm{p}(\mathcal{S}_*), \mathcal{D}) \longrightarrow \mathcal{D} \tag{4.20}$$

Definition 4.21. The *∞-category of spectra* is the category of spectrum objects in pointed spaces.

5. Generalized Cohomology Theories

We shall now use the language of ∞-categories to reformulate the concept of generalized cohomology theory à-là Eilenberg-Steenrod. In this new context, we recall a representability theorem for cohomology theories by spectrum object.

Remark 5.1. Denote by $\mathcal{S}\mathrm{et}^{\mathbb{Z}}$ the category of $\mathbb{Z}$-indexes families of sets. Given an object S and $n \in \mathbb{Z}$, denote by $\Sigma^n S$ the shifted family $(\Sigma^n S)_i = S_{i-n}$.

Definition 5.2 ([Lur17, 1.4.1.6])**.** Let $\mathcal{C}$ be a finitely cocomplete pointed ∞-category, $\Sigma_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ the induced suspension functor. A *generalized cohomology theory* is a functor $H : \mathrm{h}\mathcal{C}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}^{\mathbb{Z}}$ together with a natural isomorphism $\partial : \Sigma H \to H\Sigma_{\mathcal{C}}$ such that:

- H preserves arbitrary products. In particular, $H^n(0)$ is the one-point set. Given an object X, the unique morphism $X \to 0$ induces an element $* \simeq H^n(0) \to H^n(X)$, which we denote by 0.
- Given a coexact triangle $X' \to X \to X''$, if $\eta \in H^n(X)$ has image $0 \in H^n(X'')$, then it lies in the image of $H^n(X') \to H^n(X)$.

Theorem 5.3 ([Lur17, 1.4.1.10])**.** *Let $\mathcal{C}$ be a nice ∞-category and (H, ∂) a generalized cohomology theory, then, for every n, the functor H^n is representable by an object $E(n)$.*

The natural isomorphism ∂ translates into an equivalence $E(n) \simeq \Omega E(n+1)$, which is then used to construct a spectrum object representing the cohomology theory H^n, see [Lur17, 1.4.1.11].

Remark 5.4. For $\mathcal{C} = \mathcal{S}_*$, the above definition of cohomology theory reduces to the classical Eilenberg-Steenrod definition. Since $\mathcal{S}_*$ is nice, we thus recover the classical *Brown representability theorem*.

References

[GJ99] Paul G. Goerss and John F. Jardine. *Simplicial homotopy theory*, volume 174 of *Progress in Mathematics*. Birkhäuser Verlag, Basel, 1999.

[Gro10] Moritz Groth. A short course on ∞-categories. *arXiv preprint*, 2010. arXiv:1007.2925.

[Hir03] Philip S. Hirschhorn. *Model categories and their localizations*, volume 99 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2003.

[Lur09] Jacob Lurie. *Higher topos theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 2009.

[Lur17] Jacob Lurie. Higher algebra. Available online, September 2017.

[Lur26a] Jacob Lurie. Kerodon (Tag 00TK). https://kerodon.net/tag/00TK, 2026.

[Lur26b] Jacob Lurie. Kerodon (Tag 019D). https://kerodon.net/tag/019D, 2026.

[Lur26c] Jacob Lurie. Kerodon (Tag 01Z4). https://kerodon.net/tag/01Z4, 2026.

[Lur26d] Jacob Lurie. Kerodon (tg 0066). https://kerodon.net/tag/0066, 2026.

[Lur26e] Jacob Lurie. The Homotopy Category of an ∞-Category. https://kerodon.net/tag/0049, 2026.